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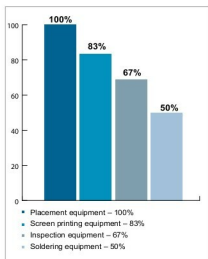


Fig. 4: Survey participants' rankings of the major demand generating product segments

flexible gantries and high-accuracy optical encoder systems are examples of product innovations as a result of technological advancements. A majority of the industry experts indicated that set-top boxes were the most demand-generating product segment.

Demand for SMT equipment is also high in telecom (including mobile handset manufacturing) and energy segments. Moreover, low-cost, lighting-specific SMT equipment have started making inroads into the burgeoning LED industry.

Equipment in demand

Placement equipment represent the largest product segment within the SMT equipment market. Experts unanimously predict a steady demand in this segment attributed to the growth of SMT assembly lines to enable efficient electronics manufacturing practices and innovation related to product miniaturisation and complex circuits. For the same

reason, soldering and printing equipment are also expected to experience decent demand (Fig. 4).

Sixty-seven per cent of the experts emphasised the importance of online quality assurance during manufacturing. They indicated that the improved speed and flexibility of quality assurance procedures had generated demand for advanced SMT inspection systems including automatic X-ray inspection (AXI) systems and automatic optical inspection (AOI) systems. AXI equipment are expected to gain from the integration of computer tomography and intuitive programming. AOI equipment are expected to become more popular due to the use of machine-vision-based algorithms that enable easy programming, flexibility and enhanced ability to detect faults quickly while minimising false call rates.

Market pulse

Industry experts believe that there are many advantages of using SMT equipment, such as lower manufacturing costs, higher component density, as well as simpler and faster automation leading to smart factories. These advantages coupled with emerging technological trends will shape the SMT manufacturing equipment market in India:

Focus on precision. Technological advancements in SMT will largely focus on bringing precision into the manufacturing process to reduce defects, while optimising the use of assembly operations to improve productivity.

Demand for one-stop solutions. There is an increased demand for integrated solutions to enable fully

automated control. Manufacturers are shifting from using individual SMT machines to integrated one-stop SMT assembly lines in order to get a competitive advantage.

Shift to 0201 geometry. The Indian industry has moved straight to the 0201 geometry, leapfrogging the phase of making larger SMT products.

Affordable equipment for small products. SMT equipment for small products are becoming affordable. This trend is already driving the market well.

Demand for small 'things'. The rapid increase in the use of smaller IoT devices (known as 'things') will drive the SMT market.

Moreover, almost all the component vendors have migrated their processes to SMT to get the advantages of increased manufacturing speed, advanced complementary metal-oxide semiconductor (CMOS) technology, and decreased interconnect widths in ICs.

Moving forward

Worldwide, the SMT equipment market is primarily driven by the EMS industry. However, this industry has not matured enough in India compared to other Asian countries. Most of the Tier 2 and Tier 3 EMS vendors still act as assemblers (doing just PCB assembly job work) rather than making the entire 'box build' unit or executing the complete production of an electronic device. Transformation of EMS players from assemblers to 'box build' service providers will not only enhance the demand for SMT manufacturing equipment but also help EMS players to gain a competitive edge in the market.

However, for this, EMS players need to develop three major capabilities: in-house supply chains (covering component engineering), design for excellence (DFX, where 'X' stands for cost, assembly, manufacturing, testing and the environment), and finally, test and repair capabilities. **EFY**